Bayesian Optimization vs. CMA-ES: Effects of Rotation and Conditioning

November 13, 2026

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Github Repository: https://github.com/DSA4212-Project23/project23

Abstract

We investigate how rotation and conditioning affect Bayesian optimization (BO) and CMA-ES on noiseless quadratic problems with known optima. An initial 500-evaluation comparison exposed frequent zero-expected-improvement fallbacks in our BO implementation. Controlled acquisition ablations and frozen-model checks identified numerical underflow and insufficient local candidate coverage. We then evaluated a fixed log-EI/local-candidate revision on three unused problem instances. At 100 evaluations, both revised BO kernels achieved lower geometric-mean regret than CMA-ES in all four conditions, but required much more computation. ARD BO retained a large rotation penalty on ill-conditioned problems. The results support an explanation involving both coordinate-sensitive modelling and acquisition implementation, with conclusions limited by the small instance sample and the restricted problem family.

1 Research question and hypothesis

Does rotating a narrow valley make it harder to optimise, and does the effect increase with conditioning? We compare two GP-based BO configurations with CMA-ES while preserving the feasible domain, objective scale and initial information between orientations. Following the emphasis on interpretable benchmark difficulties in black-box optimisation [1], we use custom problems with exact solutions; these are not official BBOB functions.

Our hypothesis is that BO with separate coordinate length scales (ARD) benefits from aligned anisotropy but suffers a larger rotation penalty as conditioning increases. An isotropic GP provides a control: its distance measure is rotation-invariant, although its finite acquisition search need not produce rotated copies of individual trajectories. CMA-ES adapts a full covariance matrix using rankings and offers a contrasting way to learn local geometry [2]. We also test whether apparent BO stagnation is caused partly by acquisition arithmetic and candidate search, rather than treating it as an intrinsic property of BO.

2 Optimisation problem

The completed comparisons use five-dimensional, noiseless quadratics:

$$F_Q(x) = \frac{1}{C}\sum_{j=1}^{d} \lambda_j z_j^2, \quad z = Q^\mathsf{T} x - a, \quad \|x\|_2 \le 1, \quad \lambda_j = \kappa^{(j-1)/(d-1)}, \qquad d = 5.$$

The matrix Q is the identity for aligned problems and a seeded random rotation otherwise. The seeded shift has $\|a\| = 0.25$. All terms are nonnegative, with unique minimiser $x^\star = Qa$ and minimum zero. The Hessian is $2Q\,\mathrm{diag}(\lambda_j)Q^\mathsf{T}/C$, so its condition number is exactly κ. We compare $\kappa = 1$ and 10000; the former is a spherical negative control.

We set

$$C = \sum_{j=1}^{d} \lambda_j \left(\frac{1}{d+2} + a_j^2\right),$$

so $\mathbb{E}[F_Q(X)] = 1$ for a uniform point in the unit ball. This follows from $\mathbb{E}[X_j] = 0$ and $\mathbb{E}[X_j^2] = 1/(d+2)$. The ball is unchanged by rotation and $F_Q(Qx) = F_I(x)$: orientation changes neither the underlying landscape nor the feasible geometry. A fixed box would confound rotation with boundary geometry. Normalisation controls scale, but does not make every condition equally difficult at a fixed target regret.

Only the evaluator accesses Q, the optimum or derivatives. The algorithms receive objective values and initial observations. The generator also supports quartic terms and other dimensions, validated mathematically but not used in the comparisons reported here (Appendix A).

3 Methods

3.1 GP models and acquisition search

BO fits a probabilistic surrogate and chooses evaluations using an acquisition function [3]. We use a constant-amplitude Matérn-$5/2$ GP. ARD learns one length scale per coordinate; the isotropic model shares one:

$$\rho_\ell(x, x') = \sqrt{\sum_j (x_j - x'_j)^2/\ell_j^2}.$$

ARD represents a diagonal metric, which generally cannot express arbitrary rotated anisotropy. The isotropic distance is preserved under joint rotation of training and query points.

Responses are normalised during fitting and predictions returned to the objective scale. Fixed diagonal regularisation is $\alpha = 10^{-8}$; it is not observation noise. Length scales start at 0.5 with bounds $[0.01, 10]$; amplitude starts at 1 with bounds $[0.001, 1000]$. Marginal-likelihood fitting uses two additional random restarts initially; subsequent fits warm-start from the previous kernel with no additional restarts. These rules are shared by every acquisition variant.

For incumbent loss b, predictive mean μ and standard deviation $s > 0$, expected improvement for minimisation is

$$\mathrm{EI}(x) = (b - \mu)\Phi(u) + s\phi(u), \qquad u = (b - \mu)/s.$$

For $s = 0$, EI is $\max(b - \mu, 0)$. Our original implementation samples 512 uniform ball candidates and refines the ten highest-EI eligible candidates with constrained SLSQP. We impose $1 - x^\mathsf{T}x \geq 0$, use gradient $-2x$, at most 100 iterations and tolerance 10^{-9}. Ordinary EI is scaled by its positive pool maximum. A point must be finite, feasible within 10^{-12} and more than 10^{-8} from previous observations. Pool candidates remain eligible when local refinement fails. When every pool EI is zero, the first eligible random candidate is selected and a fallback is recorded.

We test two changes in a 2×2 acquisition ablation. Stable log-EI uses the tail formulation of Ament et al. [5]; monotonicity preserves the mathematical EI maximiser while avoiding underflow. Mixed candidates retain 384 global points and add 64 feasible novel perturbations at each radius 0.01 and 0.1 around the observed incumbent. They use no optimum information. Both changes retain ten SLSQP starts and the original GP and constraint rules. The combined variant, `logei_local`, is the *revised BO* used in the held-out comparison.

3.2 CMA-ES and common recommendations

We use pycma 4.5.0 with active full covariance adaptation [2, 4]. CMA-ES starts at the best initial point, with covariance I, step size $0.3/\sqrt{d}$ and population $4 + \lfloor 3 \ln d \rfloor$ (eight here). Mirrors and restarts are disabled; remaining settings use the pinned library defaults. A small library diagonal perturbation is compensated to obtain the intended identity covariance. BO uses all initial observations for its GP; CMA-ES uses their best point to initialise its mean.

Outside-ball CMA proposals are resampled, consuming time but no objective calls; 10,000 rejections within a generation cause explicit failure. This preserves the domain's rotational symmetry but truncates unconstrained Gaussian sampling. An incomplete final population consumes only the remaining budget and does not update covariance. Both algorithms recommend their best evaluated feasible point, including initial points; no extra final-mean evaluation is used. Normal stopping reasons, partial failures and actual calls are recorded.

4 Experimental protocol

Table 1 separates development, mechanism testing and validation. All conditions use $d = 5$, quadratic shape, $\kappa \in \{1, 10000\}$ and both orientations, except the timing pilot, which uses only the rotated $\kappa = 10000$ case. Each run includes 11 initial observations in its budget. The original and held-out comparisons each have six runs per condition/method.

Study	Instances	Search seeds	Calls/run	Runs
Original BO / CMA	0–2	0–1	500	72
Acquisition ablation	0	0	100	32
Revised BO / CMA	3–5	0–1	100	72
Timing pilot	3	0	500	3

Table 1: Completed experiments. The ablation has four acquisition variants and two GP kernels. Other studies have ARD BO, isotropic BO and CMA-ES.

The master seed is 20261102, with separate streams for shifts, rotations, initial designs and optimiser randomness. Methods share initial points; paired orientations rotate those points with the problem. Subsequent proposals are not coupled using the hidden rotation, so shared seed integers do not imply identical rotated trajectories. Settings were fixed within each study. The ablation motivated the revised BO variant; its settings were frozen before using instances 3–5. CMA-ES stayed unchanged. Thus the held-out comparison uses new instances of the same family, not new problem classes. Neither the changed budget nor the changed cohort is treated as a controlled acquisition ablation; that evidence comes from the separate 32-run diagnostic.

Regret is $r = F_Q(\hat{x})$ because the known minimum is zero. We use $g = \log_{10}(\max(r, 10^{-12}))$, with an explicit numerical floor, and report $10^{\text{mean}(g)}$ as geometric-mean regret. Paired effects are formed before averaging search seeds within each instance. Define

$$M = g_{\text{BO}} - g_{\text{CMA}}, \quad R_\kappa = g_{\kappa,\text{rotated}} - g_{\kappa,\text{aligned}}, \quad I = R_{10000} - R_1.$$

Negative M favours BO; positive R is a rotation penalty; positive I means that penalty grows with conditioning. Exploratory 95% percentile intervals resample whole instances 10,000 times (seed 20261103), retaining both search seeds and paired conditions. Each repeated study has only three independent instances, not six. Intervals have limited resolution and no multiple-comparison adjustment. No intervals are assigned to the single-instance ablation or timing pilot.

Runs are serial with thread limits set before numerical imports. Search time includes initial evaluations, fitting, acquisition or covariance updates, constraint handling and bookkeeping. It excludes imports, problem/design creation, file output and analysis. Invalid CMA proposals cost time only. Evaluator-only checks do not feed information back to the search. Actual calls are retained, with final recommendations carried forward for normal early stops if needed; every run in these four studies reached its budget.

5 Results and explanation

5.1 Why the original BO required investigation

At 500 calls, the original BO and CMA comparison produced the results below. These concern the original EI/global-pool implementation on instances 0–2.

κ	Orientation	ARD BO	Isotropic BO	CMA-ES
1	Aligned	0.000306	0.000455	6.02e-08
1	Rotated	0.000182	0.000262	1.53e-07
10000	Aligned	2.04e-06	2.28e-05	3.87e-06
10000	Rotated	2.29e-05	5.29e-05	4.11e-06

CMA-ES achieved much lower sphere regret. ARD was best on the aligned ill-conditioned case, but rotation increased its geometric-mean regret about 11.2-fold. Its interaction was 1.277 [1.180, 1.337] log units, consistent with the coordinate-metric hypothesis. However, isotropic BO also had a positive interaction, 0.604 [0.037, 0.897], while CMA-ES's interval included zero. This suggested that GP representation alone could not explain all effects.

Moreover, ordinary-EI random fallbacks occurred on 5,109 of 11,736 ARD proposals and 4,896 of 11,736 isotropic proposals. Median search times were 43.43, 73.36 and 0.01825 seconds for ARD, isotropic and CMA-ES respectively. These implementation diagnostics made it inappropriate to attribute the BO plateaus solely to the underlying optimisation method.

5.2 Testing numerical underflow and candidate coverage

The four acquisition variants were tested on identical initial designs and fixed model settings. All eight baseline trajectories reproduced the first 100 observations from the original study. The combined revision improved final regret in all eight diagnostic cases; Appendix A retains the complete table, including counterexamples for individual changes. Full curves are available in the repository. This is one instance and one search seed per setting.

We also replayed 21 original GP states using their warm-start and random-stream history, comparing four proposals per frozen model. Before evaluation 90 in the aligned ARD sphere run, all 512 global candidates had positive predictive uncertainty but ordinary EI equal to zero. Maximum log-EI was -799.08, confirming numerical underflow. Mixed candidates exposed 128 positive-EI points. Under the unchanged GP, ordinary EI with local candidates proposed loss 1.94×10^{-6} and the combined variant 7.87×10^{-7}, versus incumbent loss 3.81×10^{-4}. These true-loss checks were offline diagnostics, never training observations in the replay.

Candidate coverage explains another part of the failure. On the sphere, $F(x) = \|x - x^\star\|^2/C$, with $C = 5/7 + 0.25^2$. The strictly improving ball at this incumbent has radius $r = \sqrt{CF(x_{\text{best}})} \approx 0.0172$ and lies inside the domain. A fresh uniform pool of 512 points hits it with probability $1 - (1 - r^5)^{512} \approx 7.7 \times 10^{-7}$. Local refinement could still reach it from outside, but tiny/zero EI can obstruct that route. The frozen-model improvement shows that the surrogate was not wholly uninformative: useful acquisition values became accessible near the incumbent.

Across eight runs per variant, the baseline used 44 random fallbacks and each revised variant used zero. Yet log-EI alone worsened the rotated ARD sphere case, and local candidates alone worsened the rotated ill-conditioned ARD case. Neither change guarantees improvement on every trajectory. Total search times across each set of eight runs were 125.6 seconds (baseline), 303.7 (log-EI), 213.3 (local) and 291.5 (combined), excluding frozen-model replay. Improved acquisition search therefore has a computational cost.

5.3 Held-out evaluation efficiency and geometry

All 72 held-out runs completed 100 calls without failures or missing records. Both BO columns below use the frozen combined revision; CMA-ES is unchanged.

κ	Orientation	ARD BO	Isotropic BO	CMA-ES
1	aligned	2.74×10^{-6}	2.50×10^{-6}	1.33×10^{-2}
1	rotated	1.65×10^{-6}	2.31×10^{-6}	1.13×10^{-2}
10000	aligned	8.66×10^{-7}	6.06×10^{-4}	2.45×10^{-3}
10000	rotated	3.82×10^{-4}	5.06×10^{-4}	4.39×10^{-3}

Both revised BO variants have lower geometric-mean regret than CMA-ES in all four conditions. All eight BO-minus-CMA intervals lie below zero. For example, ARD's difference is -3.452 $[-3.632, -3.115]$ on the aligned ill-conditioned case and -1.060 $[-1.557, -0.508]$ after rotation. Figure 1 shows evaluation efficiency over the full budget, not only the final score.

The geometry effect persists: ARD's ill-conditioned rotation effect is 2.645 [2.496, 2.861], approximately 441-fold higher geometric-mean regret. Its interaction is 2.864 [2.526, 3.205], whereas the isotropic interaction is -0.044 $[-0.141, 0.132]$ (Figure 2). This supports the coordinate-metric explanation on held-out cases after improving acquisition.

CMA-ES also has a positive interaction, 0.323 [0.226, 0.381], complicating a claim that only ARD responds to rotation. Its ill-conditioned rotation effect is 0.253 $[-0.029, 0.443]$. Such finite-sample contrasts do not establish a violation of distributional rotation invariance: later random proposals are not coupled through Q, and only three instances are represented. Sphere-control rotation intervals include zero for every method.

5.4 Computation and the separate timing pilot

At 100 calls, median search times (IQRs) were 35.87 [27.04, 42.07] seconds for ARD, 36.37 [29.53, 42.77] for isotropic BO and 0.00436 [0.00425, 0.00448] for CMA-ES. The evaluation-efficiency advantage therefore costs substantial computation on this cheap objective. These are machine-specific timings, not an equal-wall-time trial.

The separate pilot below uses only rotated $\kappa = 10000$, instance 3, seed 0. All three fresh runs completed 500 calls and exactly reproduced their corresponding held-out first-100 input/output pairs. Times are cumulative within each pilot run; no bootstrap intervals are appropriate.

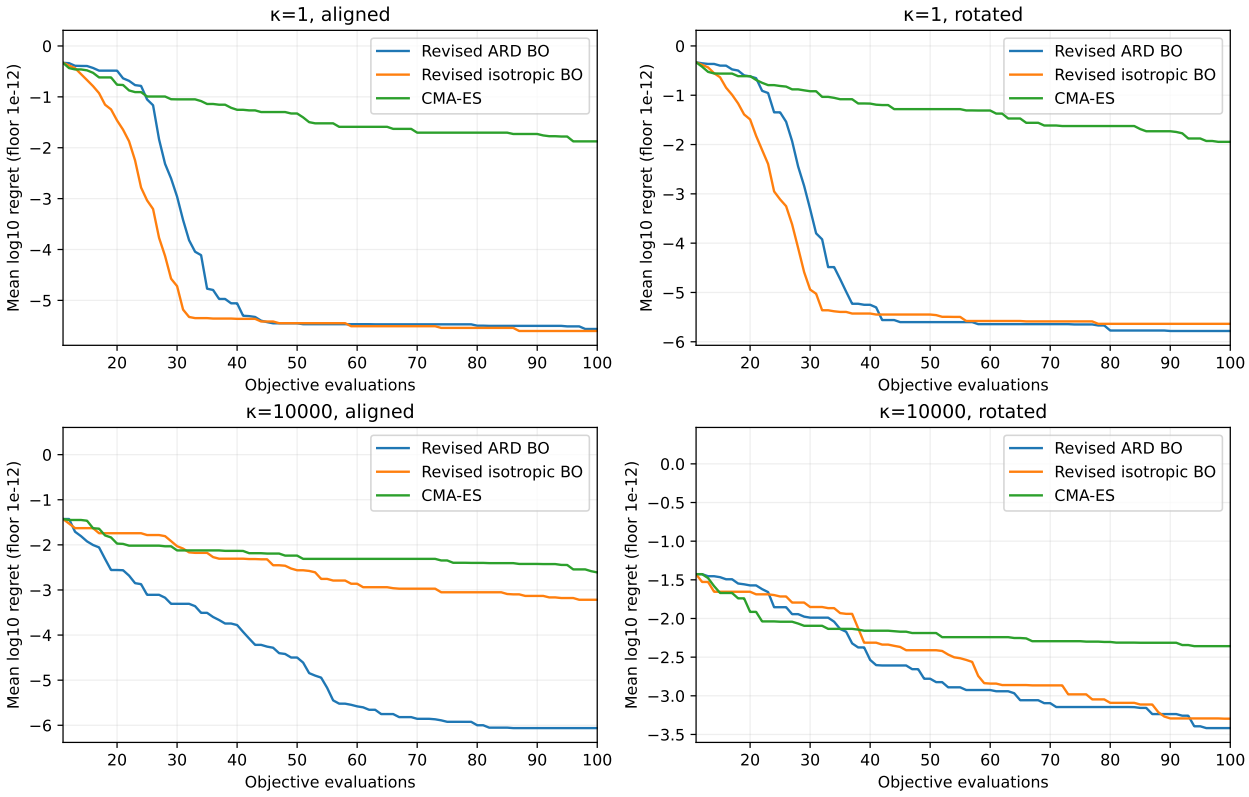


Figure 1: Held-out convergence at 100 calls. Curves average log-regret over three instances and two search seeds each; lower is better. All methods share their first 11 observations. Panels have different vertical scales. These are evaluation-budget curves, not equal-wall-time comparisons.

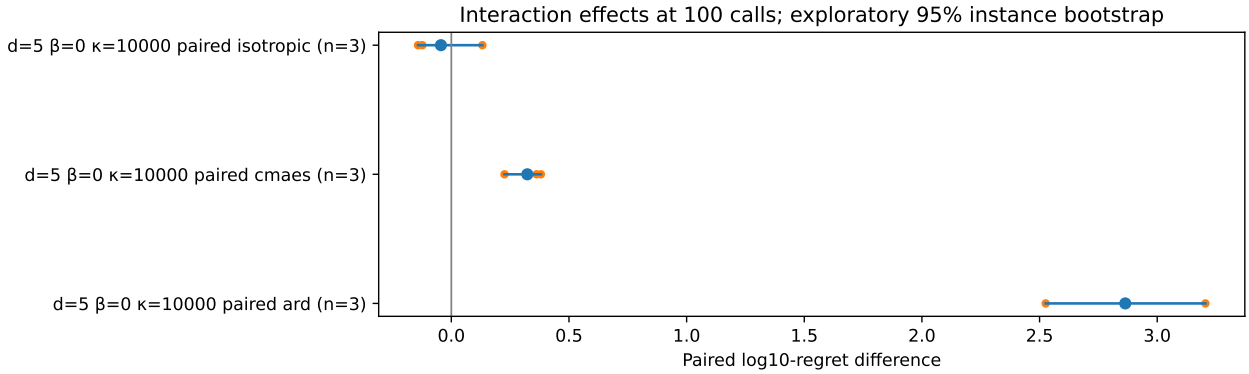


Figure 2: Increase in rotation penalty from $\kappa = 1$ to 10000 at 100 calls. ARD and isotropic labels denote revised BO. Orange dots are instance means; blue dots and lines are aggregate means and exploratory 95% instance-bootstrap intervals. Positive values indicate a larger penalty at high conditioning.

Method	Regret, 100	Regret, 500	Seconds, 100	Seconds, 500
ARD BO	5.76×10^{-4}	1.11×10^{-7}	27.44	319.43
Isotropic BO	8.95×10^{-4}	5.20×10^{-7}	24.21	280.87
CMA-ES	3.78×10^{-3}	6.30×10^{-5}	0.00493	0.01944

Five times the evaluation budget required approximately 11.6 times the elapsed time for both BO kernels and 3.94 times for CMA-ES. ARD spent 56.88 seconds fitting and 262.54 selecting candidates; isotropic BO spent 22.83 and 258.02 seconds respectively. Objective evaluation took under 0.01 seconds per run. Both BO variants retained lower regret at 500 calls here, but this single case does not establish a replicated 500-call ranking. Their late plateaus despite zero random fallbacks also show that removing one failure mechanism does not guarantee continued progress.

6 Limitations and conclusion

The experiment supports two related explanations. First, acquisition arithmetic and candidate coverage can materially limit a GP-based optimiser even when its surrogate still offers useful guidance. Second, after addressing those issues, ARD retains a much larger conditioning–rotation interaction than isotropic BO, consistent with its diagonal coordinate metric. The CMA-ES counterexample and finite instance sample prevent a stronger invariance claim.

The scope is five-dimensional, noiseless, single-basin quadratics. Three instances per repeated study give limited uncertainty estimates, and GP warnings remain frequent (Appendix A). No claim is made about noise, multimodality, high dimensions, parallel workers or general algorithm superiority. The earlier and held-out cohorts differ in both instances and budget and cannot be pooled to estimate an acquisition effect. The ablation shares an instance with development; held-out validation reduces, but does not eliminate, concerns about selecting an implementation for this family. CMA-ES did not receive a comparable acquisition-style tuning study, so the revised comparison evaluates these selected configurations, not equally tuned algorithm families.

For these tested conditions, revised BO uses a small evaluation budget well, while CMA-ES has much lower computation cost. ARD's strength on aligned narrow valleys does not transfer unchanged to rotated valleys. Algorithm choice must therefore account for geometry, acquisition implementation and the relative costs of objective calls and optimiser overhead. More instances, dimension and quartic checks, and equal-wall-time or expensive-objective experiments are useful future extensions; they are not completed evidence in this report.

References

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A Validation, ablation and diagnostics

All seven benchmark checks passed. Landscape validation covered 180 settings across dimensions 2, 5 and 10, three conditioning levels, both supported shapes and ten instances. Maximum rotation-equivalence error was 1.95×10^{-14} and relative condition-number error 1.16×10^{-12}. Quartic and dimension extensions were validated mathematically but were not part of the optimisation comparisons. Further checks cover acquisition values, feasibility, budgets, reproducibility, stopping, pairing, bootstrap and resume.

The complete ablation below reports 100-call regret on instance/run 0. "Local" uses mixed candidates; "Both" combines them with log-EI. Model settings and budgets are fixed. Additional local draws change later random streams; frozen-model audits instead clone the candidate RNG.

Kernel	κ	Orientation	Baseline	Log-EI	Local	Both
ard	1	aligned	0.000381	3.14e-05	1.98e-06	3.06e-06
ard	1	rotated	0.000155	0.000227	1.65e-06	2.15e-06
ard	10000	aligned	4.28e-06	1.04e-07	8.12e-07	1.33e-07
ard	10000	rotated	0.000254	0.00016	0.000497	4.26e-05
isotropic	1	aligned	0.000896	5.3e-06	3.69e-06	9.66e-07
isotropic	1	rotated	0.000102	4.12e-05	4.49e-06	3.64e-06
isotropic	10000	aligned	0.000579	0.000258	0.000103	0.000347
isotropic	10000	rotated	0.00062	0.000428	0.000352	0.000295

Neither revised BO kernel used random fallbacks in the held-out study (2,136 proposals each). ARD had 187 failed SLSQP exits and 2,006 warned GP fits; isotropic BO had 153 failed exits and 1,579 warned fits. Each kernel attempted 21,360 refinements. Warnings count fits with at least one warning, not failed runs. Feasible pool candidates remain eligible after refinement failure. CMA-ES rejected 86 outside-ball proposals, charging time but no calls. All 179 study runs reached their budgets; this does not mean every internal numerical optimisation converged.

The repository retains full diagnostic curves, warning records and numerical results. All code uses functions and dictionaries in `code/preparebenchmark.py`. Raw records include observations, timings, settings, versions and source fingerprints. All 75 held-out/pilot records were reused successfully in resume checks; 136 protected original artifacts retain their checksums. The README separates saved-data verification from fresh experiment reproduction. Early implementation pilots are excluded from the reported comparisons. The full study record is `report/comparison.md`.